

CSC 7740 Final Project: Team Information and Topic

Fall 2026

Project title

Identifying Coordinated Amazon Review Groups with Distributed Text and Graph Analysis

Team member names with LSU email addresses

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Prior CSC 4740 enrollment: None of the team members has previously taken CSC 4740 at LSU.

Dataset(s) you plan to use with their URLs

Amazon Reviews'23 (McAuley Lab, UC San Diego) — <https://amazon-reviews-2023.github.io/>

Component	Contents	Compressed	Raw
User reviews	24,447,530 reviews, 4 categories (Video Games, CDs & Vinyl, Arts Crafts & Sewing, Baby Products), 1996–2023	3.71 GB	12.86 GB
Item metadata	1,858,398 products: store/brand, price, category hierarchy, catalog rating and rating count	1.05 GB	4.25 GB
Total	gzip-compressed JSONL (1 GB = 10^9 bytes)	4.76 GB	17.11 GB

Direct downloads (one `.jsonl.gz` file per category):

- Reviews: https://mcauleylab.ucsd.edu/public_datasets/data/amazon_2023/raw/review_categories/
- Metadata: https://mcauleylab.ucsd.edu/public_datasets/data/amazon_2023/raw/meta_categories/

Big data framework(s) you plan to use

- **Apache Spark** (PySpark DataFrame API) — distributed data cleaning, metadata joins, window operations, and partitioned processing
- **Spark MLlib** — MinHash locality-sensitive hashing to score review-text similarity within candidate groups
- **GraphFrames** — distributed connected components on the account co-review graph
- **Spark Standalone on cloud VMs** — multi-node deployment, with Amazon S3 and Apache Parquet for input and output storage

Short description (theme) of your application

We will build a distributed Spark pipeline to identify groups of Amazon accounts that repeatedly review the same products within the same short time windows, across 24.4 million reviews and their product metadata. PySpark will clean all reviews, join them with product metadata, write Parquet files partitioned by category and year, and exclude reviews shorter than 30 characters only from MinHash text analysis. Spark window functions will create product-time candidate groups; oversized groups will be divided into smaller time buckets, and account pairs appearing together in at least three distinct product-time groups will form weighted edges in a GraphFrames graph. Connected components on this graph will identify candidate coordinated groups, ranked by group size, edge density, repeated co-review frequency, time concentration, rating agreement, deviation from each product's catalog average rating, and review-text similarity measured with MLlib MinHash. We will run the pipeline on a multi-node cloud cluster, compare one-worker and multi-worker runtimes, and use verified-purchase and helpful-vote fields to describe the results.